

Ghost Font: A Formal Specification of Motion-Encoded Anti-AI Typography

Formalized from the work of Eric Lu / Mixfont (2026)

Research analysis and mathematical specification

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1 Introduction

Ghost Font, released by Eric Lu and the Mixfont team in July 2026, is a motion-based visual encoding system that embeds a short text message into a video clip in a form immediately legible to human observers but opaque to current AI multimodal systems. Unlike traditional steganographic schemes that hide information in the spatial or spectral domain of a single frame, Ghost Font encodes information in the *temporal domain*—specifically in the relative motion direction of two overlapping dot populations. This document provides a formal mathematical specification of that encoding algorithm.

1.1 Motivation and Context

Contemporary large language models with vision capabilities process video by decomposing it into a sequence of individual frames and applying image-based analysis to each frame independently. They do not, in general, perform native temporal integration of motion across frames. The human visual system, by contrast, possesses dedicated neural machinery (the dorsal “where/how” stream, area MT/V5) for detecting and integrating motion across time. Ghost Font exploits this architectural asymmetry: the letter signal exists only in the *inter-frame motion differential*, which humans perceive instantly but which current AI frame-by-frame pipelines cannot recover without explicit motion-compensation preprocessing.

A secondary layer—a *decoy message*—is superimposed as a faint static pattern. When an AI system averages frames or scans for spatial patterns, it surfaces this decoy rather than the true message, acting as a honeypot.

2 Notation and Primitives

Definition 1 (Canvas). A canvas $\mathcal{C} = (W, H)$ is a discrete integer grid of pixels with width $W \in \mathbb{Z}^+$ and height $H \in \mathbb{Z}^+$. A pixel is addressed by (x, y) with $x \in [0, W)$, $y \in [0, H)$, all integers.

Definition 2 (Frame). A frame at time step t is a function $F_t : \mathcal{C} \rightarrow \{0, 1\}$, where 0 denotes a background (light) pixel and 1 denotes a foreground (dark) dot.

Definition 3 (Dot Population). A dot population $\mathcal{P}$ is a finite multiset of particles. Each particle p_i carries an initial position $(x_i^{(0)}, y_i^{(0)}) \in \mathcal{C}$ drawn uniformly at random, and belongs to exactly one of two classes: signal or background.

Definition 4 (Letter Mask). The letter mask $\mathcal{M} : \mathcal{C} \rightarrow \{0, 1\}$ is a binary function derived by rasterizing the message glyphs onto the canvas at a chosen font face and size. $\mathcal{M}(x, y) = 1$ iff pixel (x, y) falls within a glyph region.

3 The Ghost Font Encoding Algorithm

The algorithm proceeds in four stages: (1) canvas and mask initialization, (2) dual-population dot seeding, (3) frame generation via opposing motion, and (4) decoy superimposition.

3.1 Stage 1: Canvas and Mask Initialization

Given a message string s , rasterize each glyph onto the canvas using a chosen typeface, producing the letter mask $\mathcal{M}$. Glyphs are typically laid out horizontally centred on the canvas.

3.2 Stage 2: Dual-Population Dot Seeding

Let $\rho \in (0, 1)$ be the target dot density. Seed two disjoint populations uniformly at random:

$$\mathcal{P}_{\text{sig}} = \{p_i : (x_i^{(0)}, y_i^{(0)}) \sim \text{Uniform}(\mathcal{C}), \mathcal{M}(x_i^{(0)}, y_i^{(0)}) = 1\}, \quad (1)$$

$$\mathcal{P}_{\text{bg}} = \{p_j : (x_j^{(0)}, y_j^{(0)}) \sim \text{Uniform}(\mathcal{C}), \mathcal{M}(x_j^{(0)}, y_j^{(0)}) = 0\}, \quad (2)$$

with $|\mathcal{P}_{\text{sig}}| + |\mathcal{P}_{\text{bg}}| \approx \rho \cdot W \cdot H$.

Remark 1. *The mask determines class membership, not ongoing spatial confinement. At time $t > 0$, particles move freely across the entire canvas; a signal particle originally inside a glyph region may later be anywhere.*

3.3 Stage 3: Frame Generation via Opposing Motion

The encoding mechanism assigns opposite vertical velocity vectors to the two populations.

Definition 5 (Signal and Background Velocities). *Let $v \in \mathbb{R}^+$ be the nominal scroll speed in pixels per second, $\Delta t = 1/f_r$ the inter-frame interval for frame rate f_r , and $(w_x, w_y) \in \mathbb{R}_{\geq 0}^2$ optional wobble amplitudes with angular frequencies (ω_x, ω_y) . Then*

$$\mathbf{v}_{\text{sig}}(n) = (w_x \sin(\omega_x n \Delta t), -v + w_y \cos(\omega_y n \Delta t)) \quad (\text{upward}), \quad (3)$$

$$\mathbf{v}_{\text{bg}} = (0, +v) \quad (\text{downward, constant}). \quad (4)$$

(Upward is the $-y$ direction in standard image coordinates.)

The position of signal particle p_i at integer frame index n is

$$\begin{pmatrix} x_i(n) \\ y_i(n) \end{pmatrix} = \begin{pmatrix} x_i^{(0)} + \sum_{k=0}^{n-1} w_x \sin(\omega_x k \Delta t) \\ \left(y_i^{(0)} - n v \Delta t + \sum_{k=0}^{n-1} w_y \cos(\omega_y k \Delta t) \right) \bmod H \end{pmatrix}, \quad (5)$$

with vertical wrap-around modulo H to maintain uniform coverage. The background particle p_j satisfies

$$(x_j(n), y_j(n)) = (x_j^{(0)}, (y_j^{(0)} + n v \Delta t) \bmod H). \quad (6)$$

Each frame F_n is rendered by placing a filled disk of radius r at each particle position:

$$F_n(x, y) = 1 \iff \exists p_i \in \mathcal{P}_{\text{sig}} \cup \mathcal{P}_{\text{bg}} \text{ s.t. } \|(x, y) - (x_i(n), y_i(n))\|_2 \leq r. \quad (7)$$

3.4 Stage 4: Decoy Superimposition

Let $\mathcal{D} : \mathcal{C} \rightarrow \{0, 1\}$ be the binary raster of a decoy string d (e.g., WRITTEN IN GHOST FONT). Each output frame is

$$F_n^{\text{out}}(x, y) = \min(1, F_n(x, y) + \alpha_d \mathcal{D}(x, y)), \quad (8)$$

where $\alpha_d \ll 1$ is the decoy opacity, chosen so that the decoy is invisible in any single frame but accumulates to visibility when frames are temporally averaged (see Section 5).

Algorithm 1 Ghost Font Video Encoding

Require: Message s , canvas (W, H) , speed v , frame rate f_r , dot density ρ , wobble $(w_x, w_y, \omega_x, \omega_y)$, total frames N_f , decoy string d , decoy opacity α_d

Ensure: Video $\mathbf{V} = [F_0^{\text{out}}, \dots, F_{N_f-1}^{\text{out}}]$

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1:  $\mathcal{M} \leftarrow \text{RASTERIZEGLYPHS}(s, W, H)$ 
2:  $\mathcal{D} \leftarrow \text{RASTERIZEGLYPHS}(d, W, H)$ 
3:  $\mathcal{P}_{\text{sig}}, \mathcal{P}_{\text{bg}} \leftarrow \text{SEEDDOTS}(\mathcal{M}, \rho, W, H)$ 
4: for  $n = 0$  to  $N_f - 1$  do
5:    $F_n \leftarrow \mathbf{0}_{W \times H}$ 
6:   for each  $p_i \in \mathcal{P}_{\text{sig}}$  do
7:      $(x, y) \leftarrow \text{ADVANCESIG}(p_i, n, v, \Delta t, w_x, w_y, \omega_x, \omega_y, H)$ 
8:      $\text{PLACEDOT}(F_n, x, y, r)$ 
9:   end for
10:  for each  $p_j \in \mathcal{P}_{\text{bg}}$  do
11:     $(x, y) \leftarrow \text{ADVANCEBG}(p_j, n, v, \Delta t, H)$ 
12:     $\text{PLACEDOT}(F_n, x, y, r)$ 
13:  end for
14:   $F_n^{\text{out}} \leftarrow \min(1, F_n + \alpha_d \cdot \mathcal{D})$ 
15: end for return  $\mathbf{V}$ 
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4 Perceptual Properties

4.1 Human Legibility via Kinetic Depth and Motion Segregation

Human motion perception (*kinetic depth effect*, *structure-from-motion*) allows the visual system to segment two coherently moving dot fields even when they overlap and each individually appears random. The signal region—dots scrolling upward—segregates from the background—dots scrolling downward—making letter shapes immediately apparent to a human viewer, typically within one second of video playback.

4.2 Single-Frame Uninformativeness

Proposition 1. *For any single frame F_n produced by Algorithm 1 (ignoring the α_d decoy term), the pixel distribution is i.i.d. Bernoulli(ρ), independent of the letter mask $\mathcal{M}$.*

Proof sketch. Each particle’s initial position is drawn uniformly from $\mathcal{C}$. Advancing by a deterministic offset (modulo H) maps a uniform distribution to a uniform distribution. The union rendering of two independent uniform-density Bernoulli processes at combined rate ρ is again Bernoulli(ρ) at each pixel, with no spatial structure that depends on which population placed the dot. $\square$

Consequently, a screenshot (single frame) carries zero mutual information about the letter mask.

4.3 Message Recovery by Motion-Compensated Integration

Define the vertical shift operator $\mathcal{T}_{\delta y}[F]$ that translates frame F upward by δy pixels (with wrap-around). The signal population at frame n is displaced by $\delta y_n = n v \Delta t$ (ignoring wobble) relative to frame 0. Applying the inverse shift before summation yields

$$I_{\text{sig}}(x, y) = \frac{1}{N_f} \sum_{n=0}^{N_f-1} \mathcal{T}_{n v \Delta t}[F_n](x, y). \quad (9)$$

In this shifted coordinate, signal dots accumulate coherently while background dots are smeared uniformly across all y -positions. The result is a high-contrast image of the letter mask $\mathcal{M}$, with signal-to-noise ratio growing as $\mathcal{O}(\sqrt{N_f})$.

5 Security Analysis

5.1 Layer 1 — Single-Frame Opacity

By Proposition 1, no information about s is recoverable from any individual frame. Image-based AI analysis (screenshot, single-frame OCR) yields null information about the true message.

5.2 Layer 2 — Temporal Averaging and the Decoy Honeypot

Naïve temporal averaging without motion compensation gives

$$\bar{F}(x, y) = \frac{1}{N_f} \sum_{n=0}^{N_f-1} F_n^{\text{out}}(x, y) \approx \rho \cdot \mathbf{1} + \alpha_d \mathcal{D}(x, y). \quad (10)$$

The moving populations average out to uniform density ρ ; the only spatially structured component that survives is the *static* decoy $\alpha_d \mathcal{D}$. An AI applying frame averaging therefore recovers d , not s —the honeypot property. The model reports this finding with high confidence, further masking the existence of the true message.

5.3 Layer 3 — Motion-Compensation Break

A determined adversary with local code execution can defeat both layers:

1. Estimate the scroll velocity v via optical flow (e.g., Farnebäck or Lucas–Kanade).
2. Apply the motion-compensated integrator of Section 4.3 to reconstruct $\mathcal{M}$.
3. Run OCR on the reconstructed image.

This constitutes a complete break. The **ghost-font-breaker** tool (42-evey, GitHub, 2026) demonstrates deterministic message recovery on all publicly available Ghost Font clips with recovered velocity $v = 4.012$ px/frame, wobble amplitudes $w_x = 72$ px, $w_y = 48$ px.

5.4 Resistance Summary

Adversary capability	Method	Decoy?	True msg?
Screenshot / single frame	Direct OCR	No	No
Frame averaging	Naïve avg.	Yes	No
Online multimodal model	Frame-by-frame	Yes	No
Local code execution	Motion-compensated avg.	Yes	Yes

6 Parameter Reference

Table 1: Ghost Font algorithm parameters (observed / recovered values)

Parameter	Symbol	Observed value	Notes
Scroll speed	v	120 px/s (4.012 px/frame)	at ≈ 30 fps
Signal direction	$\mathbf{v}_{\text{sig}}$	upward ($-y$)	
Background direction	$\mathbf{v}_{\text{bg}}$	downward ($+y$)	
Wobble amplitude x	w_x	72 px	sinusoidal
Wobble amplitude y	w_y	48 px	cosinusoidal
Dot density	ρ	$\approx 0.3\text{--}0.5$	
Dot radius	r	2–3 px	
Decoy opacity	α_d	$\ll 1$	survives avg.
Decoy string	d	WRITTEN IN GHOST FONT	hardcoded
Max message length	$ s $	36 characters	prototype

7 Limitations and Future Directions

1. **Human legibility cost.** Ghost Font is harder for humans to read than ordinary text. The trade-off is inherent: motion encoding requires a video medium and reduces reading speed compared to static typography.
2. **Video-native AI models.** The current resistance relies on AI systems processing video as frame sequences. Once video-native models performing native temporal integration of motion become commonplace, the core evasion mechanism will likely fail.
3. **No cryptographic security.** Ghost Font provides obfuscation, not cryptographic secrecy. A motivated adversary with code execution can fully recover the message. True secrecy requires a shared key; a companion design document proposes key-encrypted extensions.
4. **Character limit.** The current prototype supports short messages (≤ 36 characters). Longer messages reduce per-letter motion contrast and degrade human legibility.

5. **CAPTCHA potential.** Motion-based encoding is a promising direction for next-generation CAPTCHAs, exploiting the human advantage in biological motion perception where static-image CAPTCHAs have been largely defeated by current vision models.

8 Conclusion

Ghost Font formalizes as a *dual-population, opposing-direction random dot kinematogram* in which the letter mask $\mathcal{M}$ partitions dots into a signal population (scrolling up at velocity v) and a background population (scrolling down at v). Single frames are statistically uniform (Proposition 1); the letter mask is recoverable only by motion-compensated temporal integration (Section 4.3). A static decoy layer provides a honeypot for frame-averaging adversaries. The system is provably opaque to single-frame analysis, practically opaque to current online multimodal AI, and fully defeatable by local code-execution via optical flow and motion-compensated integration—a known limitation that motivates the key-encrypted extensions described in the companion design document.